

Chapter – 2

Lenskart E-Commerce Database

2 Introduction

The entities required for the **Lenskart E-Commerce Database** are identified based on the business requirements of an online eyewear shopping platform. The attributes of each entity, along with their primary keys, foreign keys, unique keys, and attribute types, are determined to ensure efficient data organization. The relationships among the entities are identified to represent customer interactions, product management, order processing, payments, and shipment tracking. This analysis serves as the foundation for designing the Entity Relationship (ER) Diagram, relational schema, and database implementation while ensuring data integrity, consistency, and minimal redundancy.

2.1 Entity Analysis

2.2.1 Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
Customer_Name	Not Null
Email	Unique Key
Phone_Number	Unique Key
Address	Not Null

2.2.2 Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key

2.2.3 Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Brand	Not Null
Price	Not Null
Stock	Not Null
Category_ID	Foreign Key

2.2.4 Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	Not Null
Total_Amount	Not Null
Order_Status	Not Null

2.2.5 Entity: Order_Item

Attribute	Key Type
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Not Null

Attribute	Key Type
Price	Not Null

2.2.6 Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	Not Null
Payment_Status	Not Null

2.2.7 Entity: Shipment

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Tracking_Number	Unique Key
Delivery_Status	Not Null

2.2.8 Entity: Admin

Attribute	Key Type
Admin_ID	Primary Key
Admin_Name	Not Null
Email	Unique Key
Password	Not Null

2.3 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Places	Order	One-to-Many	Partial / Total
Category	Contains	Product	One-to-Many	Total / Total
Order	Includes	Order_Item	One-to-Many	Total / Total
Product	Appears In	Order_Item	One-to-Many	Partial / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Admin	Manages	Product	One-to-Many	Partial / Total
Admin	Manages	Order	One-to-Many	Partial / Total

2.4 Conclusion

The entity analysis for the **Lenskart E-Commerce Order Management System** successfully identifies the core entities, their attributes, primary keys, foreign keys, and relationships required to support the system. The analysis ensures that customer information, product details, categories, orders, payments, shipments, and administrative activities are efficiently organized within the database. By defining appropriate relationships, cardinality, and participation constraints, the database design minimizes data redundancy while maintaining data integrity and consistency. This entity analysis serves as the foundation for developing the **Entity Relationship (ER) Diagram**, relational schema, normalization process, and the overall implementation of the Lenskart E-Commerce Database Management System.